H4: Grafen (uigewerkte oplossingen)

Commando voor het bepalen van de "row reduced echelon form"

```
(%i1)   rref(A):=block([p,q,k],[p,q]:matrix_size(A),A:echelon(A),
            k:min(p,q),
            for i thru min(p,q) do (if A[i,i]=0 then (k:i-1,return())),
            for i:k thru 2 step -1 do (for j from i-1 thru 1 step -1 do A:rowop(A,j,i,A[j,i])),
            A)$
```

```
(%i2)   columndiv(A,i,a):=columnop(A,i,i,1-1/a);
```

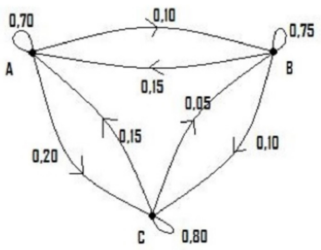
(%o2) $\mathrm{columndiv}(A,i,a):=\mathrm{columnop}\left(A,i,i,1-\frac{1}{a}\right)$

1) Oef 5

1.1 (a)

Opgelet:
(*) de graaf en matrix geven de verandering weer na 5 jaar.
(*) C=Z

Figure 1:



(%i3)
```
M: matrix(
 [0.7,0.15,0.15],
 [0.1,0.75,0.05],
 [0.2,0.1,0.8]
);
```

(M) $$\begin{bmatrix} 0.7 & 0.15 & 0.15 \\ 0.1 & 0.75 & 0.05 \\ 0.2 & 0.1 & 0.8 \end{bmatrix}$$

(%i4)
```
B: matrix(
 [0.2],
 [0.3],
 [0.5]
);
```

(B) $$\begin{bmatrix} 0.2 \\ 0.3 \\ 0.5 \end{bmatrix}$$

1.2 (c)

Situatie na 5 jaar (dus éénmaal M laten inwerken op de beginsituatie B):

(%i5)
```
M.B;
```

(%o5) $$\begin{bmatrix} 0.26 \\ 0.27 \\ 0.47 \end{bmatrix}$$

Situatie na 10 jaar (dus tweemaal M laten inwerken op de beginsituatie B):

(%i6)
```
(M^^2).B;
```

(%o6) $$\begin{bmatrix} 0.2929999999999999 \\ 0.252 \\ 0.4550000000000001 \end{bmatrix}$$

1.3 (c)

Stabiele situatie:

We lossen het stelsel (M-I)X=0 op
Dit correspondeert met de uitgebreide matrix M-I | 0
Bovendien voldoen x, y en z aan de vergelijking x+y+z=1

Opmerking: we geven hier onmiddellijk de matrix M-I in (+ extra vergelijking)
en vertrekken niet van het stelsel (dit zou veel meer typwerk vragen).

(%i7)
```
MIX: matrix(
 [0.70-1,0.15,0.15,0],
 [0.1,0.75-1,0.05,0],
 [0.2,0.1,0.8-1,0],
 [1,1,1,1]
);
```

(MIX)
$$\begin{bmatrix} -0.3 & 0.15 & 0.15 & 0 \\ 0.1 & -0.25 & 0.05 & 0 \\ 0.2 & 0.1 & -0.2 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

```
(%i8)    rref(MIX);
0 errors, 0 warnings
rat: replaced -0.3 by -3/10 = -0.3
rat: replaced 0.15 by 3/20 = 0.15
rat: replaced 0.15 by 3/20 = 0.15
rat: replaced 0.1 by 1/10 = 0.1
rat: replaced -0.25 by -1/4 = -0.25
rat: replaced 0.05 by 1/20 = 0.05
rat: replaced 0.2 by 1/5 = 0.2
rat: replaced 0.1 by 1/10 = 0.1
rat: replaced -0.2 by -1/5 = -0.2
```

(%o8)
$$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{3} \\ 0 & 1 & 0 & \frac{2}{9} \\ 0 & 0 & 1 & \frac{4}{9} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Opmerking:
4e rij = nulrij => het stelsel is niet strijdig, dus de gezochte stabiele situatie bestaat!

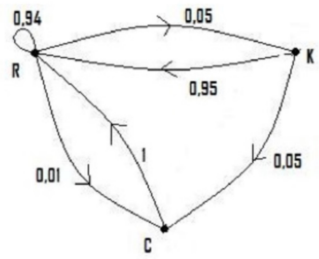
Antwoord:
De markt evolueert naar 33,33% voor merk A, 22,22% voor merk B en 44,45% voor merklozen computers.
(We ronden het laatste percentage af naar boven om som 100% te verkrijgen.)

2) Oef 6

2.1 (a)

R = in roulatie
K = in klein onderhoud
G = in groot onderhoud

Figure 2:



(%i9)
```
M: matrix(
 [94/100,19/20,1],
 [1/20,1/20,0],
 [1/100,1/20,0]
);
```

(M)
$$\begin{bmatrix} \frac{47}{50} & \frac{19}{20} & 1 \\ \frac{1}{20} & \frac{1}{20} & 0 \\ \frac{1}{100} & \frac{1}{20} & 0 \end{bmatrix}$$

Stabiele situatie: M-I | X en x+y+z=1700

(%i10)
```
MIX: matrix(
 [94/100-1,19/20,1,0],
 [1/20,0-1,0,0],
 [1/100,1/20,0-1,0],
 [1,1,1,1700]
);
```

(MIX)
$$\begin{bmatrix} -\frac{3}{50} & \frac{19}{20} & 1 & 0 \\ \frac{1}{20} & -1 & 0 & 0 \\ \frac{1}{100} & \frac{1}{20} & -1 & 0 \\ 1 & 1 & 1 & 1700 \end{bmatrix}$$

(%i11)
```
rref(MIX);
```

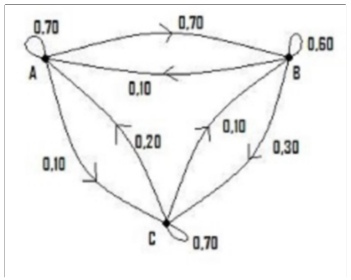
(%o11)
$$\begin{bmatrix} 1 & 0 & 0 & 1600 \\ 0 & 1 & 0 & 80 \\ 0 & 0 & 1 & 20 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

```
Antwoord:
Rijden -> 1600
Klein onderhoud -> 80
Groot onderhoud -> 20
```

3) Oef 7

3.1 (a)

Figure 3:



(%i12)
```
M: matrix(
 [0.7,0.1,0.2],
 [0.2,0.6,0.1],
 [0.1,0.3,0.7]
);
```

(M) $$\begin{bmatrix} 0.7 & 0.1 & 0.2 \\ 0.2 & 0.6 & 0.1 \\ 0.1 & 0.3 & 0.7 \end{bmatrix}$$

3.2 (b)

(%i13)
```
B: matrix(
 [0.30],
 [0.20],
 [0.50]
);
```

(B) $$\begin{bmatrix} 0.3 \\ 0.2 \\ 0.5 \end{bmatrix}$$

(%i14)
```
[M.B,(M^^2).B,(M^^3).B];
```

(%o14) $$\left[\begin{bmatrix} 0.33 \\ 0.23 \\ 0.44 \end{bmatrix}, \begin{bmatrix} 0.342 \\ 0.248 \\ 0.41 \end{bmatrix}, \begin{bmatrix} 0.3462 \\ 0.2582 \\ 0.3956 \end{bmatrix}\right]$$

3.3 (c)

(%i15)
```
MIX: matrix(
  [0.7-1,0.1,0.2,0],
  [0.2,0.6-1,0.1,0],
  [0.1,0.3,0.7-1,0],
  [1,1,1,1]
);
```

(MIX)
$$\begin{bmatrix} -0.3 & 0.1 & 0.2 & 0 \\ 0.2 & -0.4 & 0.1 & 0 \\ 0.1 & 0.3 & -0.3 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

(%i16)
```
rref(MIX);
```

```
rat: replaced -0.3 by -3/10 = -0.3
rat: replaced 0.1 by 1/10 = 0.1
rat: replaced 0.2 by 1/5 = 0.2
rat: replaced 0.2 by 1/5 = 0.2
rat: replaced -0.4 by -2/5 = -0.4
rat: replaced 0.1 by 1/10 = 0.1
rat: replaced 0.1 by 1/10 = 0.1
rat: replaced 0.3 by 3/10 = 0.3
rat: replaced -0.3 by -3/10 = -0.3
```

(%o16)
$$\begin{bmatrix} 1 & 0 & 0 & \frac{9}{26} \\ 0 & 1 & 0 & \frac{7}{26} \\ 0 & 0 & 1 & \frac{5}{13} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(%i17)
```
float(%);
```

(%o17)
$$\begin{bmatrix} 1.0 & 0.0 & 0.0 & 0.3461538461538461 \\ 0.0 & 1.0 & 0.0 & 0.2692307692307692 \\ 0.0 & 0.0 & 1.0 & 0.3846153846153846 \\ 0.0 & 0.0 & 0.0 & 0.0 \end{bmatrix}$$

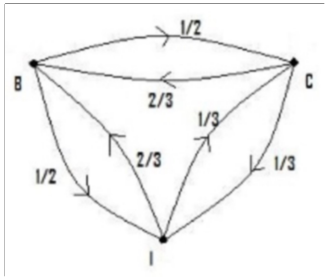
Antwoord: De markt evolueert naar 34,62% voor merk A; 26,92% voor merk B en 38,46% voor merk C.

4) Oef 8

B = Belgische keuken
C = Chinese keuken
I = Italiaanse keuken

4.1 (a)

Figure 4:



(%i18)
```
M: matrix(
 [0,2/3,2/3],
 [1/2,0,1/3],
 [1/2,1/3,0]
);
```

(M)
$$\begin{bmatrix} 0 & \frac{2}{3} & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & 0 \end{bmatrix}$$

(%i19)
```
M^^2;
```

(%o19)
$$\begin{bmatrix} \frac{2}{3} & \frac{2}{9} & \frac{2}{9} \\ \frac{1}{6} & \frac{4}{9} & \frac{1}{3} \\ \frac{1}{6} & \frac{1}{3} & \frac{4}{9} \end{bmatrix}$$

(%i20)
```
B: matrix(
 [1],
 [0],
 [0]
);
```

(B)
$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

(%i21) `(M^^2).B;`

(%o21) $$\begin{bmatrix} \frac{2}{3} \\ \frac{1}{6} \\ \frac{1}{6} \end{bmatrix}$$

Antwoord: 2/3 kans dat hij overmorgen Belgisch eet

4.2 (b)

Stabiele situatie: M-I | 0

(%i22)
```
MIX: matrix(
 [0-1,2/3,2/3,0],
 [1/2,0-1,1/3,0],
 [1/2,1/3,0-1,0],
 [1,1,1,1]
);
```

(MIX) $$\begin{bmatrix} -1 & \frac{2}{3} & \frac{2}{3} & 0 \\ \frac{1}{2} & -1 & \frac{1}{3} & 0 \\ \frac{1}{2} & \frac{1}{3} & -1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

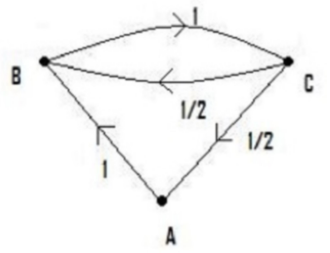
(%i23) `rref(MIX);`

(%o23)
$$\begin{bmatrix} 1 & 0 & 0 & \frac{2}{5} \\ 0 & 1 & 0 & \frac{3}{10} \\ 0 & 0 & 1 & \frac{3}{10} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Antwoord:
40% B (dus het vaakst)
30% C
30% I

5) Oef 9

Figure 5:



5.1 (a)

(%i24)
```
M: matrix(
 [0,0,1/2],
 [1,0,1/2],
 [0,1,0]
);
```

(M)
$$\begin{bmatrix} 0 & 0 & \frac{1}{2} \\ 1 & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix}$$

5.2 (b)

(%i25)
```
B: matrix(
 [1],
 [0],
 [0]
);
```

(B)
$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

(%i26)
```
(M^^2).B;
```

(%o26)
$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Antwoord: Chris krijgt na 2 worpen zeker de bal (kans = 100%)

5.3 (c)

(%i27)
```
C: matrix(
 [0],
 [0],
 [1]
);
```

(C) $$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

(%i28)
```
(M^^2).C;
```

(%o28) $$\begin{bmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

Antwoord: 50% kans dat de bal na 2 worpen opnieuw bij Chris is.

6) oef 10

6.1 (a)

(%i29)
```
M: matrix(
 [0,0.5,1.2,2.5],
 [0.6,0,0,0],
 [0,0.5,0,0],
 [0,0,0.5,0]
);
```

(M) $$\begin{bmatrix} 0 & 0.5 & 1.2 & 2.5 \\ 0.6 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0.5 & 0 \end{bmatrix}$$

6.2 (b)

(%i30)
```
B: matrix(
  [3000],
  [2500],
  [1500],
  [1000]
);
```

(B)
$$\begin{bmatrix} 3000 \\ 2500 \\ 1500 \\ 1000 \end{bmatrix}$$

Na 1 dag:

(%i31)
```
M.B;
```

(%o31)
$$\begin{bmatrix} 5550.0 \\ 1800.0 \\ 1250.0 \\ 750.0 \end{bmatrix}$$

Na 2 dagen:

(%i32)
```
(M^^2).B;
```

(%o32)
$$\begin{bmatrix} 4275.0 \\ 3330.0 \\ 900.0 \\ 625.0 \end{bmatrix}$$

Na 10 dagen:

```
(%i33) (M^^10).B;
```

(%o33)
$$\begin{bmatrix} 4790.279924999999 \\ 2913.52275 \\ 1428.9615 \\ 701.2563749999999 \end{bmatrix}$$

6.3 (c)

```
(%i34) MIX: matrix(
         [0-1,0.5,1.2,2.5,0],
         [0.6,0-1,0,0,0],
         [0,0.5,0-1,0,0],
         [0,0,0.5,0-1,0],
         [1,1,1,1,1]
       );
```

(MIX)
$$\begin{bmatrix} -1 & 0.5 & 1.2 & 2.5 & 0 \\ 0.6 & -1 & 0 & 0 & 0 \\ 0 & 0.5 & -1 & 0 & 0 \\ 0 & 0 & 0.5 & -1 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

```
(%i35) rref(MIX);
rat: replaced 0.5 by 1/2 = 0.5
rat: replaced 1.2 by 6/5 = 1.2
rat: replaced 2.5 by 5/2 = 2.5
rat: replaced 0.6 by 3/5 = 0.6
rat: replaced 0.5 by 1/2 = 0.5
rat: replaced 0.5 by 1/2 = 0.5
```

(%o35)
$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Laatste rij: 0=1, dus we hebben hier te maken met een strijdig stelsel.

Interpretatie/antwoord: een stabiele situatie bestaat niet (vliegenplaag)